

Certification CCNA Cisco Season 3



In this set of lab-based demonstrations, you are a network engineer tasked with deploying Layer 2 security on Cisco switches.

You will explore how to:

1. Configure and explore the different Cisco port security options (Protect, Restrict, Shutdown)
 - **Lesson 1** : Introduction to layer 2 security
 - **Lesson 2** : Configuration for Static Protect
 - **Lesson 3** : Configuration Restrict Dynamic
 - **Lesson 4** : Configuration for shutdown sticky
2. Configure and verify DHCP snooping and Dynamic ARP Inspection (DAI)
 - **Lesson 5** : Configure and verify Dynamic ARP Inspection (DAI)
3. CCNA Quiz

Section 1 : Securing the Switch with Essential Layer 2 Protections

Lesson 1 - Introduction to layer 2 security

Layer 2 device hardening is essential to ensure that your network infrastructure is protected against unauthorized access, MAC spoofing, ARP poisoning, and potential attacks like CAM table overflow or DHCP starvation.

Port security limits and identities:

1. MAC addresses on switch ports.
2. DHCP snooping prevents unauthorized DHCP servers from assigning IP addresses, protecting against rogue DHCP attacks.
3. Dynamic ARP inspection (DAI) validates ARP packets to prevent man-in-the-middle attacks by ensuring IP-to-MAC address bindings are correct.

Together, these three tools help protect Layer 2 devices from common threats, thereby enhancing the overall security posture of your network infrastructure.

Related CCNA v1.1 exam topic:

- **5.7 Configure and verify Layer 2 security features (DHCP snooping, dynamic ARP inspection, and port security)**

In this lab, you will explore how to:

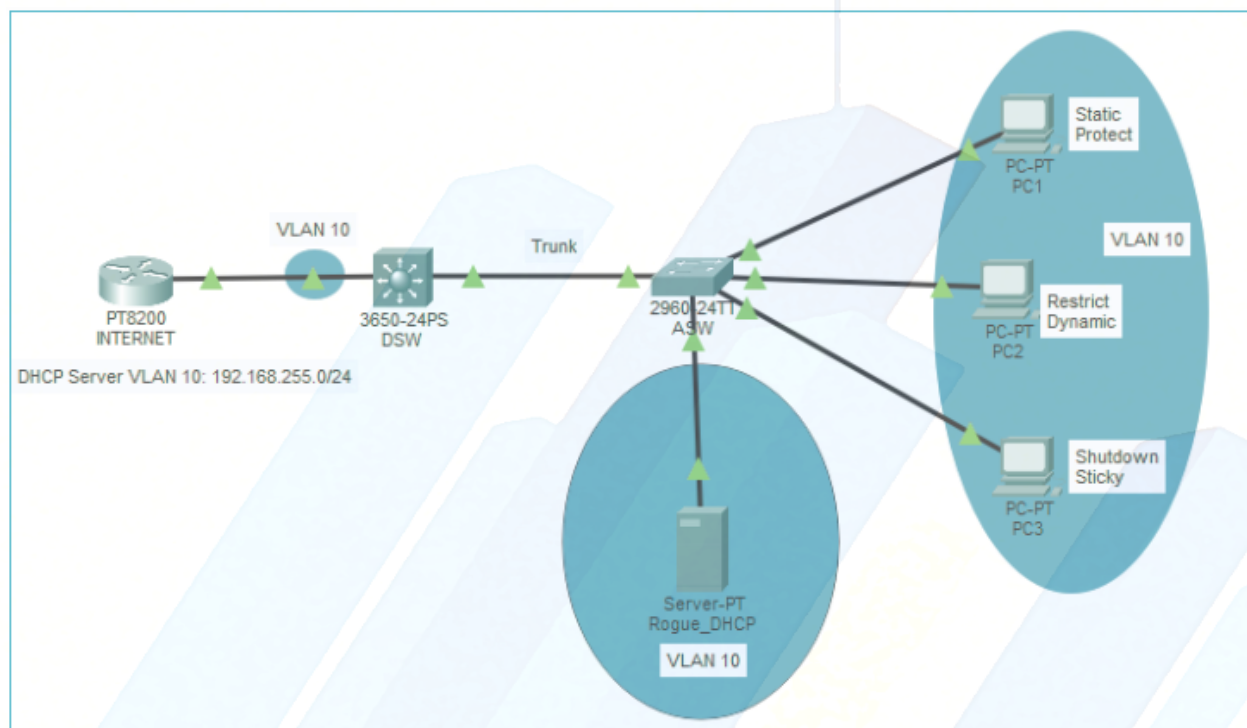
1. Configure and explore the different Cisco port security options (Protect, Restrict and Shutdown)
2. Configure and verify DHCP snooping
3. Configure and verify DAI

Notice: Before turning on Dynamic ARP Inspection (DAI), you must configure first DHCP snooping

Part 1: Setup and Scenario on Packet Tracer or CML

Notice: Refer to the topology for network configuration on Packet tracer. Before we jump into configuring any Layer 2 security features, let's explore the current network configurations.

Follow the **show run** commands on different devices and build your packet tracer network configurations. Some output has been omitted for simplicity



There are network devices to configure, the Router (INTERNET), the Distribution Switch (DSW) and the Access Switch (ASW).

Part 2: Initial Configurations

Step 1 : Configure the INTERNET Router following the show run command

```
conf t
hostname INTERNET
interface Loopback0
ip address 8.8.8.8 255.255.255.255
exit

interface gigabitethernet0/0/0
ip address 192.168.255.1 255.255.255.0
no shutdown
exit

ip dhcp pool CCNAPREP
network 192.168.255.0 255.255.255.0
default-router 192.168.255.1
```

Step 2: Configure the Distribution switch DSW following the show run command

```
conf t
hostname DSW
interface GigabitEthernet1/0/1
switchport mode trunk
exit

interface GigabitEthernet1/0/2
switchport access vlan 10
switchport mode access
```

Step 3: Configure the Access Switch ASW following the show run command

```
conf t
hostname ASW
interface gigabitethernet0/1
switchport mode trunk
exit

interface range FastEthernet0/1-3
switchport access vlan 10
switchport mode access
spanning-tree portfast
exit

interface FastEthernet0/4
switchport access vlan 10
switchport mode access
```

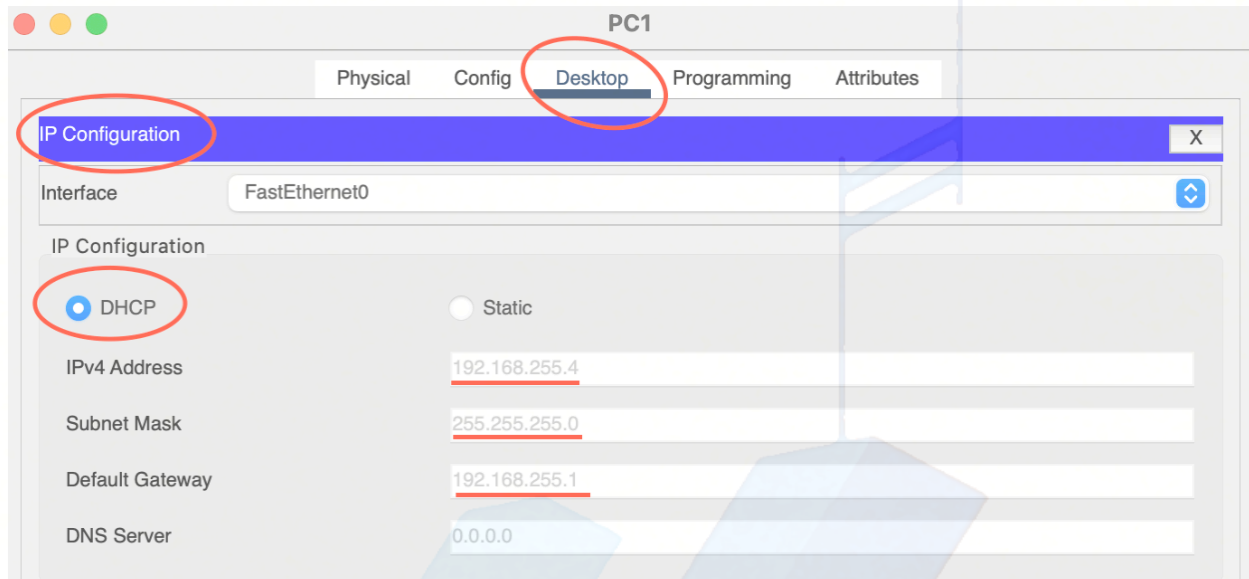
Step 4 : DHCP Request on PC1, PC2, PC3

Follow the steps to access in PC1 settings and turn DHCP request. As the **INTERNET Router** is already configured as DHCP server, the request of IP address coming the PC1 to the INTERNET Router should permits PC1 to get an IP address.

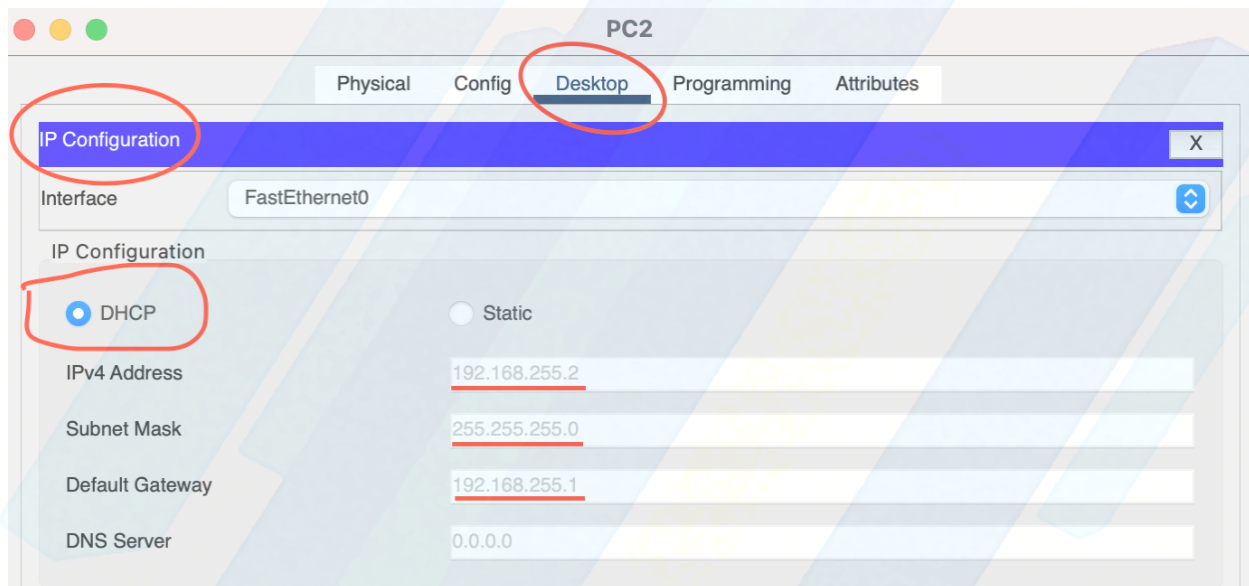
- **Click on PC1 - Go to Desktop - Go to IP Configuration - Click DHCP** to enable DHCP request

Do the same on PC2 and PC3

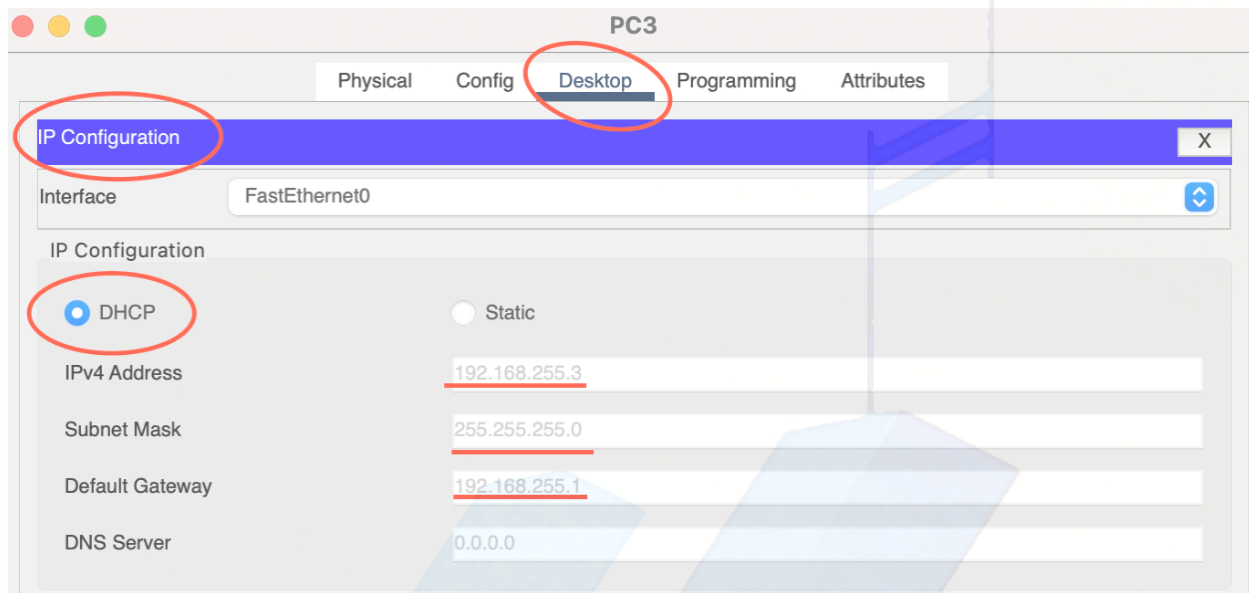
- Enable DHCP request on PC1



- Enable DHCP request on PC2

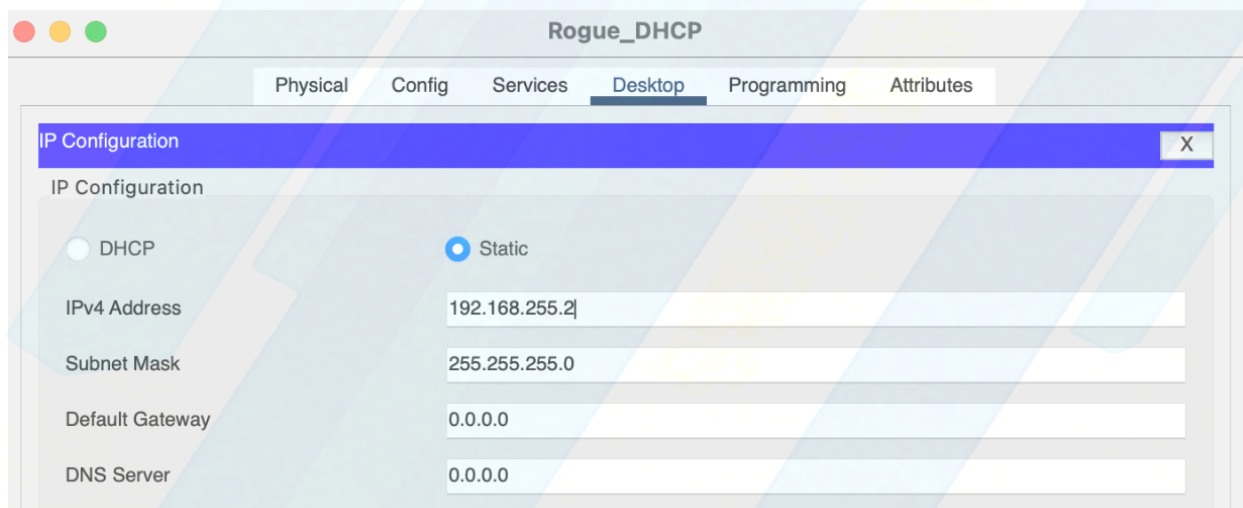


- Enable DHCP request on PC3



Step 5: Configure the DHCP service on the Rogue DHCP server

You need to set an IP address to the server first, follow the same steps as with the PC but the IP address has to be static.



To turn on DHCP on the server follow these steps : Click on the server - Go to "**Services**" - Go to "**DHCP**" and configure the DHCP parameters by these blanks:

- Pool Name
- Default Gateway
- DNS Server (Not mandatory for DHCP to work)
- Starting IP address
- Subnet Mask
- Maximun Number of users

And click on **"Add"** to add theses configurations to the DHCP Server.

Notice: This is a Rogue DHCP server so don't turn it ON for now keep it OFF to avoid DHCP conflicts with INTERNET Router which is our DHCP server.

The screenshot shows the 'Rogue_DHCP' configuration window. The 'Services' tab is selected, and 'DHCP' is highlighted in the left sidebar. The DHCP configuration form is displayed with the following settings:

- Interface: FastEthernet0
- Service: ☐ On, ☒ Off
- Pool Name: CCNAPREP
- Default Gateway: 192.168.255.2
- DNS Server: 0.0.0.0
- Start IP Address: 192.168.255.25
- Subnet Mask: 255.255.255.0
- Maximum Number of Users: 25
- TFTP Server: 0.0.0.0
- WLC Address: 0.0.0.0

Buttons: Add, Save, Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	192.168...	255.255...	231	0.0.0.0	0.0.0.0
CCNAPREP	192.168...	0.0.0.0	192.168...	255.255...	25	0.0.0.0	0.0.0.0

Next lesson 2: Configuration for Static Protect